

Literature Positioning and Novel Contributions of the Boundary-First Completion Framework

1. Positioning Statement (Problem & Gap in Paradigms)

The **boundary-first completion framework** targets the problem of inferring dynamical laws (Markov chains or semigroups) directly from *observational data*, treating the existence of an underlying model as a hypothesis to be tested rather than assumed. In complex systems and time-series, one often has reliable *finite-lag statistics* (e.g. one-step transition counts or empirical joint distributions) but no guaranteed simple mechanistic model. Traditional paradigms each fall short here: **mechanistic modeling** assumes a complete interior model (e.g. known state variables and equations) from the outset, which is often untenable in high complexity systems or leads to over-committed assumptions that data cannot justify. **Stochastic modeling** (e.g. classic time-series or HMM approaches) usually presupposes a Markov property or parametric form and fits it, which can mislead if the process isn't truly Markov of the assumed order or if multiple distinct models explain the observations equally well. **Operator-theoretic frameworks** (Koopman/Perron-Frobenius theory) elegantly linearize dynamics in infinite dimensions ¹ ², but they generally *start* from an existing dynamical system or assume a single "true" operator – they do not provide criteria for when a finite set of empirical transition statistics can be embedded into a valid semigroup or how to handle ambiguous embeddings. **Information-geometric (IG) approaches** bring powerful tools (e.g. Fisher metrics, divergence measures) to analyze statistical manifolds ³, yet classical IG is mostly developed for static distributions or parametric families, not for the *closure problem* of extending partial observations to full dynamic models. In summary, no existing paradigm fully resolves the **Markov completion** problem tackled here: *when can one reliably "complete" boundary data into a consistent dynamical model, and what if one cannot?* The boundary-first framework positions this as an *empirical regularity test*: it treats the dynamic interior as something to be *earned* via observable consistency conditions, rather than assumed ⁴ ⁵. This fills a critical gap left by mechanistic, statistical, operator, and IG paradigms by embracing the "necessary emptiness" between data and model ⁵ – i.e. acknowledging that without additional regularity, a unique or any dynamical law may *not* exist, and that this underdetermination should be characterized rather than ignored. By doing so, the framework synthesizes reductionist and complex-systems viewpoints: it recovers traditional dynamical structure when possible, but when not (or not uniquely), it provides principled diagnostics, invariants, and comparative measures instead of forcing a single spurious model ⁶ ⁵.

2. Closest Literature Clusters and Their Gaps

Below we identify **eight key clusters** of prior literature most relevant to this framework. For each cluster, we list canonical references, summarize their achievements, and note where they fall short relative to the boundary-first completion approach.

Cluster 1: Operator-Theoretic Dynamics (Koopman & Semigroup Methods)

- **Canonical references:** B.O. Koopman (1931) – introduced the Koopman operator for dynamical systems; H. Koopman & von Neumann (1932); P. Lasota & M.C. Mackey (1994) – Perron-Frobenius

operators in chaos; I. Mezić (2013) – modern Koopman spectral theory survey; M. Budisić, R. Mohr, I. Mezić (2012) – Koopman analysis tutorial ¹ ² .

- **What they achieve:** This line of work represents nonlinear dynamics through *linear operators* on function spaces (Koopman acting on observables, Perron–Frobenius on densities). It provides a powerful spectral framework: one can analyze eigenvalues, continuous spectra, and modal decompositions of dynamics ⁷ . Notably, the Koopman formalism ensures no information is lost in lifting to an infinite-dimensional linear evolution ⁸ . The approach unifies stability and ergodic analysis under a linear algebra lens, and with the Hille–Yosida theorem it connects generator spectra to long-term behavior ² . Recent advances have proven Koopman spectral decompositions for broad classes of systems and introduced practical algorithms like Dynamic Mode Decomposition (DMD) to approximate these operators ² ⁹ .
- **Shortfalls relative to our framework:** The operator-theoretic paradigm usually **assumes** an underlying dynamical system (often deterministic or Markov) is given, from which these operators are derived ⁴ . It doesn’t address the *existence problem* in reverse: given finite-time statistical summaries (e.g. an empirical transition matrix), *can* we obtain a valid one-parameter semigroup? Standard Koopman/DMD approaches will always yield *some* linear operator fit, but not all such fits correspond to a legitimate Markov process (e.g. they might not be positive or Markovian). There is little guidance on detecting when an empirical one-step operator can be “completed” into a true semigroup (satisfying Chapman–Kolmogorov consistency or Hille–Yosida conditions). In short, this literature provides the analytic language (spectra, generators) the framework builds on, but **does not solve the inference/closure question** – it presupposes the interior dynamics and doesn’t handle *partial observability* or *non-uniqueness*: if multiple different systems produce the same operator on observables, classical Koopmanism has no mechanism to highlight that equivalence (it treats one underlying system at a time). The boundary-first framework, by contrast, uses an *observables-only operator* (“*neutral operator*”) that is representation-invariant ¹⁰ and focuses on testing the semigroup property from data, addressing exactly the gap in standard Koopman theory.

Cluster 2: Delay-Coordinate Embedding and State-Space Reconstruction

- **Canonical references:** F. Takens (1981) – delay embedding theorem ¹¹ ; T. Sauer, J. Yorke, M. Casdagli (1991) – embedding for fractal attractors; N. Packard et al. (1980) – empirical attractor reconstruction; H. Whitney (1936) – mathematical embedding lemma (precursor).
- **What they achieve:** This classical nonlinear time-series theory addresses when one can *reconstruct an equivalent state space from observations*. Takens’ theorem showed that for generic smooth systems, a sufficiently high-dimensional delay vector $(y(t), y(t+\tau), \dots, y(t+(k-1)\tau))$ is a one-to-one image of the true state (for $k > 2d$, with d the attractor dimension) ¹¹ . In practice, delay embeddings allow *topologically faithful* reconstruction of deterministic dynamics: the reconstructed phase space preserves trajectories and invariant properties up to smooth coordinate changes ¹² . This is foundational in chaos theory and has become standard in nonlinear time-series analysis ¹³ . Modern extensions integrate embedding with spectral analysis: e.g. combining delays with DMD or Koopman techniques (Hankel-DMD) can recover the correct eigenfunctions and eigenvalues of the underlying system in the infinite-data limit ¹⁴ . Thus, embedding methods bridge observed sequences to an equivalent *deterministic* model, and recent results show delay coordinates make hidden spectral structure observable ¹⁵ .
- **Shortfalls relative to our framework:** Classical embedding theory assumes an *underlying deterministic system with a unique trajectory structure*. It guarantees a diffeomorphic reconstruction **if** the system is deterministic and noise-free, but it does not readily apply to stochastic dynamics or partial observability where the process might not be strictly Markov of finite order. The boundary-

first framework deals with *probabilistic* transitions (Markov chains or processes), not just deterministic flows, and it emphasizes *measure-theoretic consistency* (matching distributions) rather than just topological conjugacy. In embedding, once an attractor is reconstructed, one still typically **assumes** there is a well-defined dynamical system generating it. Our framework instead treats the step from delay-correlations to a Markov semigroup as a conditional step – requiring additional *regularity conditions* (stochastic consistency, limits) to yield a Markov process ¹⁶. Moreover, embedding theory doesn't confront the possibility of *multiple different models* producing the same delay plots or attractor geometry – it focuses on existence of *some* embedding. The completion framework extends this by asking: given one-step or finite-lag *statistical* information, when can we embed it into a *stochastic state-space model*? This is closer to a “**Markov extension problem**” which classical Takens-style results do not answer. In short, delay embedding ensures **qualitative** recovery of dynamics (assuming determinism), whereas our framework demands **probabilistic consistency** and highlights when even qualitative embeddings do not lead to a unique or valid Markov model. (Notably, the framework's hierarchy of model fidelity explicitly separates topological, measure, and geometric consistency ¹⁷ ¹⁸, and embedding addresses primarily the topological level.)

Cluster 3: Hidden Markov Models (HMMs) and State-Space Realization Theory

- **Canonical references:** D. Blackwell & L. Koopmans (1957) – identifiability of functions of a Markov chain ¹⁹; L.E. Baum & T. Petrie (1966) – foundational HMM formulation; B. Schumitsch, B. Vanluyten et al. (2007) – equivalence of HMM state representations ²⁰; L.R. Rabiner (1989) – tutorial on HMMs; K.P. Murphy (2012) – textbook on Dynamic Bayesian Networks.
- **What they achieve:** HMM theory provides a *stochastic state-space model* where an unobserved Markov chain generates observable emissions. It has been hugely successful in modeling sequences in speech, bioinformatics, etc. The literature has established algorithms for inference (Baum-Welch EM algorithm, filtering), and a theory of minimal realizations: there are conditions under which a hidden Markov process has a *unique minimal state representation up to isomorphism*. For example, Blackwell showed early on that different internal Markov chains can produce the same observable process (non-uniqueness), and later work formalized criteria for when an HMM can be reduced or when two HMMs are equivalent (e.g. having the same predictive distribution for all future sequences) ²¹. The concept of **observable (or predictive) equivalence** arises: two distinct state models are observationally indistinguishable if they define the same joint distribution over outputs ²² ²³. This cluster thus provides the basic language of partially observed Markov processes and demonstrates both the power and the pitfalls (identifiability issues) of assuming a latent Markov structure.
- **Shortfalls relative to our framework:** Classical HMM approaches typically *assume a fixed model order or number of hidden states* and then estimate parameters. They **do not treat the existence of an HMM as conditional** – the model form is a premise. The boundary-first framework flips this around: it starts from data (e.g. one-step joint distributions) and asks whether *any* Markov model exists that is compatible. If traditional HMM methods are given data violating Markov assumptions (say the process has long memory or is a mixture of models), they will still return *some* fitted HMM, potentially mis-specified, rather than flagging “no valid Markov completion.” Our framework explicitly highlights when completion fails or is non-unique, whereas standard HMM fitting lacks a formal notion of “no solution” aside from goodness-of-fit metrics. Moreover, while the HMM literature acknowledges non-identifiability (many models can produce the same output distribution), it often resolves this by selecting a minimal (smallest state) model as *the* representation. In contrast, we treat the collection of all possible realizations (the *completion groupoid*) as an object of interest, seeking invariants across that entire equivalence class rather than arbitrarily picking one (see

Novelty (iii) below). Finally, HMM realization theory is largely algebraic (based on probability transitions and linear algebra of observable matrices), whereas our framework enriches this with **information geometry** (using divergence and Fisher-metric structure on the space of transition probabilities) to gain new insights (most HMM literature does not leverage IG structure, so it misses geometric diagnostics that the boundary-first approach provides ²¹).

Cluster 4: Predictive State Representations and Behavioral Approaches

- **Canonical references:** M. Littman, R. Sutton, S. Singh (2002) – Predictive State Representations (PSRs) ²⁴ ; H. Jaeger (2000) – Observable Operator Models (OOMs); J.P. Crutchfield & B. McNamara (1987) / J.P. Crutchfield & K. Young (1989) – ϵ -machines and computational mechanics; C.R. Shalizi & J.P. Crutchfield (2001) – causal state minimality; J. Willems (1989) – behavioral systems theory (model-free viewpoint).
- **What they achieve:** These approaches all aim to *construct a state representation directly from data* without assuming a particular model form. **PSRs** (and the related OOMs) showed that one can represent a dynamical system by a state vector of **predictive probabilities**: essentially, the state is a concise summary of the distribution of future observations given the past ²⁴ . This yields a model that is *learned from observable sequences* alone, often by analyzing the Hankel matrix of future-conditioned-on-past probabilities and extracting its rank (which corresponds to the number of predictive states). Similarly, **computational mechanics** (Crutchfield et al.) introduced ϵ -machines, which identify “causal states” as equivalence classes of pasts that yield the same distribution of futures. This is effectively a minimal hidden state construction from data, guaranteeing an optimal prediction with the least internal states. **Behavioral system theory** (Willems) in control takes a related philosophy: it views a system as the set of all trajectories that satisfy certain constraints, rather than assuming an explicit state-space – thus one infers laws from data patterns (especially in linear systems, via Hankel matrices of impulse responses). The achievement of this cluster is a paradigm shift: it *doesn't start with latent variables*, it starts with observed data and builds the notion of state or dynamics as a *summary of observations*. These methods have successfully been applied to learn models when classical assumptions (like fully observed Markov states) don't hold, and they emphasize that models are not unique – one seeks a *minimal sufficient representation* instead ²⁵ .
- **Shortfalls relative to our framework:** The boundary-first framework is philosophically aligned with this cluster – indeed it also insists on using only “edge-level” observable statistics as primitives. However, PSR/OOM methods and computational mechanics don't incorporate the *notion of a closure test with respect to a family of divergences or limiting behaviors*. They will always produce a predictive model of some order (up to model complexity limits and statistical error), but they don't give clear conditions for when a *true Markov chain of any finite order exists*. In practice, one chooses a model dimension by heuristics (e.g. thresholding singular values in a Hankel matrix). By contrast, our framework defines explicit conditions (consistency across scales, α -divergence tests, etc.) that *must be met* for a genuine Markov completion to exist, and it treats failure of those conditions as informative (indicative of complex or non-Markov structure). Another difference is the **role of geometry**: PSRs are mostly algebraic/linear in flavor, whereas we use an information-geometric lens to define equivalences and distances between realizations. For example, the *neutral empirical operator* in our framework uses an f -divergence to remain “model-neutral” ^{10 26} , something not present in standard PSR/OOM formalisms. Finally, while predictive approaches do acknowledge many models can fit the same data, they typically focus on finding one minimal model (like the ϵ -machine). The completion framework instead explicitly frames the collection of all such models as a *groupoid of completions*, allowing analysis of common invariants without committing to a single realization. In short, predictive state methods provide algorithms to build models from data (and are

direct antecedents), but they don't offer a *structured theory of when this process succeeds or fails* under varying epistemic stances (α) – a gap our framework fills by introducing diagnostic conditions and tunable divergence-based criteria for model validity.

Cluster 5: Information Geometry – Foundations (Fisher Metric & α -Divergences)

- **Canonical references:** I. Csiszár (1967) – f -divergences introduction; M. Ali & D. Silvey (1966) – divergence measures; S. Amari (1985) – *Differential-Geometrical Methods in Statistics* (Fisher information geometry); O. Čencov (1972) (transl. 1982) – proof of Fisher metric's uniqueness; S. Amari & H. Nagaoka (2000) – *Methods of Information Geometry* (textbook) ²⁷.
- **What they achieve:** This rich literature treats families of probability distributions as differentiable manifolds endowed with the **Fisher information metric** and related structures. A key result (Čencov's theorem) is that the Fisher-Rao metric is essentially unique under mild invariance requirements (it is the only Riemannian metric invariant under sufficient-statistic mappings/Markov morphisms) ²⁸. Amari built on this to define the dual pair of affine connections corresponding to the α -family of divergences: any Csiszár-Morimoto f -divergence induces the same Fisher metric but different dual connections (e.g. $\alpha = \pm 1$ give the exponential and mixture connections, $\alpha = 0$ gives the Hellinger geometry) ²⁹. Thus, a one-parameter family of information geometries is available, interpolating between likelihood-focused and moment-focused perspectives. These tools have been widely used to study statistical inference, optimal estimators, and the geometry of parametric models. In short, this cluster provides the “canonical toolkit” for measuring distances on probability distributions and for ensuring our constructions are invariant under relabeling of states or other representation changes ³⁰. The concept of **α -neutrality** (that certain IG structures do not depend on the specific α chosen) is established in IG theory ²⁶ ³¹.
- **Shortfalls relative to our framework:** Classical IG, by itself, is **not concerned with dynamical systems or partial observations** – it is mostly developed for static models or full-data likelihood settings. For example, having a Fisher metric on the manifold of *all* one-step distributions $P(Y \times Y)$ is standard, but using it to analyze *when a Markov semigroup exists* is new. The existing IG literature does not tackle the problem of extending one distribution to a consistent time-series – that requires bridging IG with operator theory and time-evolution, which is precisely the novelty of our approach ³². Moreover, while the α -divergence family is known, treating α as an “**epistemic stance**” parameter (i.e. a tunable way to emphasize different aspects of model mismatch) is a novel interpretation. Typically, one fixes α (often $\alpha = 1$ corresponding to KL divergence) for an inference task; the idea of *choosing* α based on diagnostics of completion stability or empirical properties has not been mainstream. Our framework extends IG by applying it in new contexts (e.g. the *neutral geometry* on the space of transitions, IG foliations of model families, etc.) and by introducing new invariants and hypotheses specific to dynamics (e.g. divergence-zero “Rose” condition corresponds to identical one-step laws ³³). In summary, the foundations of IG are a **direct prerequisite** and inspiration, but they don't alone solve our questions – we had to combine them with dynamical constructs and develop new results (like IG-leaf spectral invariants) that go beyond standard IG usage.

Cluster 6: Information Geometry in Dynamics and Time-Series

- **Canonical references:** S. Amari (1987) – IG of linear systems (transfer functions) ³⁴; S. Amari (2001) – IG of partially observed Markov chains (dependency decomposition) ³⁵; Y. Ohara, T. Tanaka, S. Fujimoto (1990s) – IG for control systems and signal processing ³⁶; N. Nakahara & S.

Amari (2002) – IG of ARMA models; **N. Ay (2001–2015)** – information theory of complex systems (e.g. multi-time mutual information); **G. Nishiyama et al. (2010)** – filtering as information projection; Recent: **T. Matsuda (2019)** – information geometry of dynamical systems on graphs ³⁷ .

- **What they achieve:** This cluster applies IG concepts to *dynamic* or *sequential* settings, partially bridging the static–dynamic gap. For instance, Amari in the 1980s introduced a Riemannian metric and dual connections on the manifold of linear system transfer functions ³⁴ , treating ARMA model families as statistical manifolds (with power spectra as coordinates) ³⁸ . Similarly, subsequent work by Ohara, Tanaka, and Fujimoto applied Fisher information metrics to state-space model classes and developed natural gradient algorithms for adaptive filtering ³⁶ . In the context of Markov processes, Amari (2001) examined the manifold of joint probability distributions of a Markov chain of increasing length (a “hierarchical manifold”) and defined an “orthogonal decomposition” of higher-order dependencies by projecting joint distributions onto submanifolds representing lower-order interactions ³⁵ . This gave an IG perspective on separating pairwise, triplewise, etc., correlations in time-series data. More recent contributions look at **non-equilibrium processes** as flows on distribution manifolds (treating time-evolution like a trajectory in probability space) ³⁹ . For example, some studies treat the Fokker–Planck equation or population dynamics as gradient flows in the space of distributions with dual-flat structures on densities vs. fluxes ⁴⁰ . In summary, this cluster shows that many dynamic phenomena can be analyzed with IG: Markov transition matrices can be seen as points on a manifold (often an exponential family) ⁴¹ ; one can project a complex process onto simpler families to quantify information loss at different time scales ⁴² ; and even filtering (state estimation) can be viewed as an IG projection in belief space. This literature establishes that **IG and dynamics are compatible** and sets precedents for some ideas (e.g. “Markov morphisms” as information-loss maps ⁴³ , or fixing lower-order marginals defines a manifold “leaf” ⁴⁴ ⁴⁵).
- **Shortfalls relative to our framework:** While these works merge IG with dynamics, they generally do so within *specific model classes* or by assuming a *full set of parameters*. For example, Amari’s analysis of Markov chains (2001) assumes the chain exists and looks at its joint distribution manifold – it does not ask “given a single-step distribution, can we obtain the chain?” but rather analyzes dependency structure if we have the chain. Our framework, in contrast, uses IG to *answer existence and uniqueness questions*: e.g. we define **IG-equivalence** of different realizations and foliate the space of models into “**IG leaves**” of observationally equivalent dynamics ⁴⁶ ⁴⁷ . That concept – that one can have a foliation where each leaf is a set of models with the same finite-dimensional marginals – is hinted at in prior work (fixing lower-order marginals defines a submanifold ⁴⁸) but we take it further by conjecturing that *all models in one leaf share certain spectral invariants* (a new idea). In general, earlier IG-in-dynamics studies either remain theoretical (with no direct testable criteria for data) or stick to parametric cases (linear Gaussian systems, etc.). Our framework is more *non-parametric* and *constructive*: it provides algorithms and conditions (like the “Rose condition” for identical edge distributions ³³ , or empirical limit tests for semigroups) to *empirically evaluate* if data sits nicely on an IG manifold that permits a completion. Previous IG-dynamics literature did not articulate something like a “boundary vs. interior” distinction or an iterative closure test – those are novel. Moreover, none of the prior works interpret Amari’s α -divergence in terms of *stance for model completion* – that is unique to our program (others used α simply as one of many metrics, not as an epistemological knob). Thus, the existing IG+dynamics literature provides important building blocks and motivation, but the boundary-first framework’s synthesis (boundary data + IG + operator theory + explicit completion criteria) and several specific constructs (completion groupoid, α -stances, IG-leaf invariants) were not present in earlier works ⁴⁹ ⁵⁰ .

Cluster 7: Divergence-Based Inference and Robust Statistics

- **Canonical references:** A. Basu et al. (1998) – density power divergence for robust estimation; T. Eguchi (1992) – estimating equations via divergence; S. Amari (2010) – α -divergence applications; A. Cichocki & S. Amari (2010) – α - β divergence family (survey); T. Minka (2005) – α -divergence in Bayesian inference (tech report).
- **What they achieve:** This line of research explores how using different **information divergences** (beyond the usual likelihood/KL) affects statistical estimation and inference. Basu et al. introduced the *density power divergence*, effectively an α -divergence, to balance efficiency and robustness: by tuning a parameter, one can down-weight outliers and model mis-specification, achieving robustness at some cost to efficiency. Similarly, various works have shown that choosing an appropriate divergence (Hellinger, KL, reverse-KL, etc.) for fitting models can mitigate issues like heavy-tailed noise or outlier contamination. Minka’s work and others in machine learning context (e.g. β -VAE, α -divergence variational Bayes) treat α as a continuum between modes and means – adjusting α emphasizes different parts of the distribution (mode-seeking vs coverage) ⁵¹ ⁵². Overall, this cluster establishes that the **choice of divergence is a subjective “stance”** about what discrepancies matter: e.g. forward KL ($\alpha \rightarrow 1$) penalizes missing mass, while reverse KL ($\alpha \rightarrow -1$) penalizes outlier points, and $\alpha=0$ (Hellinger) gives a symmetric focus. They also provide methods to select divergences or tune parameters based on desired criteria (e.g. minimize mean-squared error or maximize breakdown point in estimation).
- **Shortfalls relative to our framework:** These ideas are usually applied in *static parameter estimation* or distribution fitting, not in testing the coherence of a dynamic model. Our framework repurposes them in a novel way: we use α -divergences to define the **“neutral operator”** and to measure discrepancies between empirical multi-step transitions and what a Markov model would predict ⁵³ ³³. In doing so, we interpret α as controlling an *empirical stance on model fidelity*: for example, a smaller α (more Hellinger-like) might treat small probability mismatches as less severe, thus being more forgiving to model approximation errors, whereas $\alpha = 1$ (KL) is very strict on matching likelihoods. This becomes a way to probe **completion stability**: if the existence of a semigroup is sensitive to the divergence chosen (e.g. only exists for some α but not others), that reveals something about the data’s structure. Such an interpretation (linking α to “which errors matter” in dynamic closure) is new – prior robust inference didn’t have the notion of “model completion fails under KL but succeeds under Hellinger,” because they weren’t looking at dynamical consistency at all. Additionally, the framework proposes **practical diagnostics for selecting α** based on data characteristics (e.g. test the smoothness or noise in transitions, perhaps choosing α that yields stable estimated generators) ⁵¹ ⁵². This goes beyond the literature where α is usually chosen *a priori* or via cross-validation at best; here it’s tied to an epistemic criteria of what constitutes a meaningful deviation. The risk that our use of α -divergences overlaps with prior robust statistics is low, because the *context is entirely different*: we’re not just doing parameter estimation under heavy tails – we’re testing the existence of a Markov process, which wasn’t on the radar of those works. However, we leverage their insight that divergence choice influences outcomes, imbuing our framework with a tunable lens to examine model fit (something missing in binary yes/no classical tests). In sum, divergence-based inference gives us the tools and vocabulary (α , robustness, information loss measures), but applying it to dynamic *closure problems* and viewing α as an “empirical stance” for completion is a novel extension.

Cluster 8: Spectral Invariants and Model Equivalences in Dynamics

- **Canonical references:** J.G. Kemeny & J.L. Snell (1976) – *Finite Markov Chains* (lumpability and eigenvalues); P. Deuffhard & M. Weber (2005) – metastable Markov sets (eigen-spectrum analysis); C. Meyer (1989) – perturbation of Markov chains eigenvalues; B. Vanluyten, B. De Moor, J. Willems (2007) – HMM minimality and equivalence ²¹ ; J. Shalizi & C. Moore (2003) – common information in hidden processes; I. Mezić (2005) – spectral factorization of dynamics.
- **What they achieve:** These works deal with the question: when two different models (or reduced models) share certain properties, especially spectral characteristics. In Markov chain theory, **lumpability** results tell us that if a chain's states can be partitioned such that the lumped process is Markov, the *aggregated chain* shares some of the original chain's eigenvalues (particularly the slow eigenmodes related to long-term behavior are preserved under proper lumping) ⁵⁴ ⁵⁵ . More generally, isospectral theory in dynamics shows that distinct systems can have identical spectra – e.g. different graphs can be isospectral. In linear systems, two minimal realizations of the same transfer function have identical pole spectra by definition. The Willems/Vanluyten line specifically looked at **equivalence of HMMs**: conditions under which one can transform one hidden Markov model into another without changing the observed process, and how to algorithmically reduce an HMM to a minimal form (often via a form of state merging) ⁵⁶ . Some works (Shalizi, Moore) also discuss measures of shared information or predictive features among different models of the same data, hinting at invariants beyond just likelihood. In summary, this cluster establishes that certain aspects of dynamics – notably spectral properties like eigenvalues or long-time scales – can be invariants of an *equivalence class* of models. For Markov processes, the notion that the **spectrum of the transition operator** (excluding perhaps transient eigenvalues tied to specific state distinctions) might depend only on the observational data (not the particular realization) is plausible from these results ⁵⁷ . For example, if two Markov chains are observationally equivalent, they necessarily have the same stationary distribution and the same leading eigenvalue (1.0 for Markov, plus subdominant eigenvalues governing mixing rates) for the observable dynamics.
- **Shortfalls relative to our framework:** What's been lacking is a **general and constructive identification of invariants across non-unique models**, especially in non-linear or non-parametric settings. Prior results are often limited to special cases – e.g. lumpability deals with exact aggregation of states, HMM equivalence criteria tend to assume finite-state and give algorithms but not easily generalizable invariants. Our framework introduces the idea of a **“completion groupoid”** of all models realizing the same boundary data and posits that within an IG leaf (i.e. an equivalence class of models sharing the same one-step distribution and other empirical traces), certain spectral quantities (like the eigenvalues of the associated neutral operator or the resolvent) are **invariant** ⁵⁸ ⁴⁷ . This is essentially a new hypothesis/generalization inspired by both IG and operator theory. While linear system theory trivially supports this (same transfer function $\Rightarrow$ same poles), for nonlinear or stochastic systems this is *not proven* in general – in fact, this is one of the framework's conjectures or proposed theorems (the “resolvent spectral conjugacy within IG leaves” claim) ⁵⁸ ⁴⁷ . No prior literature has a result like “if two distinct systems lie in the same information-geometric equivalence class, their Koopman generators are spectrally conjugate” – this is exactly one of the novel contributions (if validated). Thus, the framework goes beyond known invariants (like some shared eigenvalues under lumping) by suggesting a deeper equivalence: effectively, *all* spectral properties measurable from observables might coincide for models in the same leaf. It also proposes practical ways to use this: e.g. one could compare spectral estimates from different reconstruction methods (say delay-embedding vs. an HMM fit) to confirm they lie on the same IG leaf ⁵⁰ . The risk of overlap with existing ideas is low here, because it's a cutting-edge conceptual claim – at most, it resonates with known facts in linear systems or trivial cases (where it's obviously

true), but extending it to broad classes of nonlinear dynamics is new and partly conjectural. We are explicit that some aspects (like full **IG-leaf spectral conjugacy**) are conjectural and to be proven with further theoretical work or empirical evidence. Prior art gives us *analogies* (like lumping invariants, spectral subspace comparisons ⁵⁹ ⁶⁰), but not a general framework – our work is making that leap.

3. Novelty Matrix: Key Contributions vs Prior Art

Below is a matrix summarizing each major **novelty claim** of the boundary-first framework, against the closest prior work and how our approach differs. We also note differing assumptions, how each claim could be validated, and the risk of overlap with existing ideas.

Novelty Claim	Closest Prior Art	What's New Here	Different Assumptions	How to Test / Validate	Risk of Overlap
(i) Boundary-first stance via edge marginals – Treat empirical one-step/two-step distributions as fundamental “edges,” only derive a model if earned.	<i>Predictive State Representations</i> (Littman et al. 2002) and <i>Observable Operator Models</i> (Jaeger 2000) – state is constructed from data, not assumed ²⁴ ²⁵ . <i>Behavioral systems theory</i> (Willems 1989) – model defined by observed trajectories rather than latent equations.	We center the entire inference on the empirical stepwise marginal (the “edge”) and explicitly require any model to <i>produce</i> those edges exactly. Unlike PSR/OOM, which always yields some model, we have a built-in notion of <i>failure or non-uniqueness</i> as a meaningful outcome (no single model assumed if data doesn’t warrant it). We integrate this stance with information geometry (neutral divergence criteria) to remain representation-invariant ³⁰ , which previous approaches did not do.	No presupposed latent state space or Markov order (PSRs often assume a finite predictive state dimension exists; we do not until tested). We assume only that one can measure consistent one-step probabilities; prior approaches often assume infinite data and an underlying Markov process. Our framework is “agnostic” and neutral about the existence of a model, whereas earlier methods typically presume one <i>does</i> exist and try to find it.	Apply the framework to data from a known system and see if and when it “refuses” a model. For example, generate sequences from a complex process that violates Markov assumptions, and show that our criteria detect the violation (no valid completion), whereas a standard HMM or PSR would try to fit a model regardless. Conversely, with data from a true Markov chain, show that the framework successfully finds a completion. This tests the stance of not forcing a model: does it correctly leave things “unexplained” when it should?	Low-Moderate. The philosophy of “let the data speak” is shared by PSR and behavioral schools, so the general stance isn’t unique. However, our formalization (explicit boundary→interior conditions, IG-based neutrality) is new. There’s little risk someone has already done exactly this, but reviewers might say PSRs already <i>sound</i> similar. The defense is that we offer a rigorous <i>framework</i> with conditions and consequences for failing those conditions, which PSRs don’t have.

(ii) Completion as closure (Markov completion as a regularity earned, not assumed) – Posing the existence of a Markov chain/semigroup as a conditional closure problem on empirical data.

Kolmogorov extension theorem (1933) – gives conditions for extending consistent finite-dimensional distributions to an infinite process (but not specifically Markov). *Markov chain embedding problem* (e.g. Israel, Rosenthal & Wei 2001) – whether a given stochastic matrix is embeddable in continuous time. *System identification* (Ljung 1999) – fits models to data but assumes a model form a priori.

We explicitly formulate **criteria (compatibility and limits) under which a collection of finite-lag marginals “closes” into a full Markov process** ¹⁶ ⁵ . This treats dynamic law inference as a *subset of Kolmogorov extension with extra Markov constraints*. The novel contribution is identifying *practical regularity conditions* (e.g. our “Rose condition” for identical one-step distributions ³³ , plus multi-step consistency and a continuum limit for generators) that *guarantee* a Markov chain or semigroup exists ⁶¹ ⁶² . Unlike previous work, we frame these as empirical tests: e.g. check if $\{\gamma_Y(\Delta t)\}$ satisfy Chapman–

We assume *nothing* beyond observed marginal distributions and mild consistency; others assume a particular model class (e.g. finite-state Markov, or existence of a continuous generator). We consider even the possibility that no finite-dimensional Markov representation exists (which classical identification would treat as model misspecification, not as an allowed outcome). We also allow non-uniqueness (multiple completions) as a natural situation, whereas classical approaches often implicitly assume uniqueness (or they pick one via additional criteria).

Empirically, one can gather stepwise marginals at increasing time resolution Δt from a system and test our limit conditions for a continuous-time generator (e.g. does $\{\gamma_Y(\Delta t) - I\} / \Delta t$ converge to a stable generator as $\Delta t \rightarrow 0$?). This would validate the *closure to a semigroup*. Alternatively, simulate two-step or three-step transition probabilities and see if a single-step Markov chain can reproduce them. Our framework would apply a closure test (are the two-step distributions just the one-step kernel squared?), flagging failures where traditional fitting might force a chain. A concrete validation: using synthetic data where ground truth is known (say a process that is *not* Markov of order 1 but maybe of order 2) and confirming the

Low. While the *concept* of requiring consistency for extension is classical (Kolmogorov’s consistency), applying it specifically to Markov modeling and calling out the need for a “closure” step is novel in framing. Prior art on the embedding problem is technical and limited to checking if a given matrix is an exponential of another – it doesn’t formulate a general program of treating model existence as conditional. The risk is more that this could be seen as a reinterpretation of well-known conditions (Chapman–Kolmogorov) rather than something brand new. We mitigate that by emphasizing our new criteria and the philosophical shift (model as hypothesis, not assumption).

Novelty Claim	Closest Prior Art	What's New Here	Different Assumptions	How to Test / Validate	Risk of Overlap
		Kolmogorov-like relations across scales. If they do, completion is possible; if not, that failure is informative rather than just noise. This "earned regularity" perspective is new in that most pipelines <i>assume</i> some Markov property and then proceed, whereas we require data to demonstrate it.		framework correctly indicates "no 1-step Markov completion" (unless you include second-order edges).	

(iii) **Completion groupoid (non-uniqueness and model equivalence framing)** – Organizing all possible realizations of the same observational data as an algebraic structure (groupoid), and focusing on invariants under those transformations.

Hidden Markov Model equivalences (Blackwell 1957; Vanluyten et al. 2007) – recognized that many different HMMs can generate the same observable process ²¹. *Minimal realization theory* (control theory, Kalman 1960s) – choosing a canonical representative (e.g. minimal state) for equivalence classes. *Behavioral approach* (Willems) – doesn't privilege any particular realization.

The new contribution is explicitly **treating the space of all compatible completions as a groupoid** (each model is an object, and a morphism is essentially an isomorphism or encoding between models that preserves the observed “boundary” behavior). Rather than just acknowledging non-identifiability, we provide a *framework to work within it*: e.g. define IG-equivalence (models that live on the same information-geometric leaf of equal one-step statistics) ⁴⁶, and then seek quantities that are constant on that leaf. We don't force a minimal or unique model, but instead embrace *irreducible pluralism* as an outcome ⁶. The groupoid

We assume that *not* having uniqueness is normal and meaningful – prior art often adds assumptions to regain uniqueness (e.g. minimality, generic position, distinct eigenvalues, etc.). We drop those and instead allow each equivalence class to possibly contain infinitely many models (for instance, a hidden state can be split or combined in infinitely many ways if it doesn't change the output law). Our approach doesn't require choosing one representative; it assumes analysts might deliberately work with the whole class (via invariants). This is a different mindset: others assume there is an underlying truth (even if

We can validate this perspective by demonstration: construct two very different models (say, an HMM with 2 hidden states and another with 3 hidden states) that produce exactly the same distribution of outputs. Show mathematically they are in our completion groupoid (related by a many-to-one state mapping that preserves the one-step marginal). Then use our framework's invariants (e.g. check that their edge divergence to the empirical data is zero, and perhaps their spectral signatures align). Another test: if the framework computes something like an information-geometric curvature or an invariant statistic from just the data, that statistic should match for both models. This would illustrate that focusing on the groupoid (the class) yields quantities that

Moderate. The fact of HMM non-uniqueness is well-known, so claiming “multiple models exist” isn't novel. The novelty lies in *how we handle it*. The risk is a reader saying “people already know about equivalent HMMs; minimal realization theory covers that.” We counter that by highlighting that instead of enforcing minimality or uniqueness, we leverage the whole equivalence: our use of groupoid language and pursuit of invariants is new. Category-theoretic framing of this problem is not present in standard literature, so overlap is mostly conceptual (the underlying equivalence relation is known, but our exploitation of it is new).

viewpoint is borrowed from category theory and hasn't been applied to HMMs widely – it implies one can navigate between different realizations via transformations (some invertible, some not fully if we allow partial mappings). Our framework suggests using this to understand partial comparability: e.g. two different models may both project to the same observed chain (they are in the same groupoid orbit). By packaging this properly (perhaps with a composition law for model transformations), we can reason about families of models rather than one “true” model. This is a significant departure from the traditional approach of picking a single canonical model.

unidentifiable, one usually imagines one “true” HMM that generated the data). We treat “true model” as not well-defined in such cases – only the equivalence class is.

individual models disagree on, but invariants they agree on (and indeed our method finds those). Additionally, run our completion algorithm or neutral operator on data and see if it can reveal when multiple distinct state realizations fit equally well (e.g. via detecting a flat direction in divergence or a non-unique factorization).

(iv) α -divergence as stance parameter (linked to completion stability) – Interpreting the choice of divergence (within the Csizsár–Amari α -family) as selecting which discrepancies between empirical data and model are treated as significant, and relating this to whether a stable completion exists.

Robust statistics / density power divergence (Basu 1998) – introduced a tunable α -like parameter to balance fit vs. robustness in estimation. *Variational inference divergences* (Minka 2005; Higgins 2017 β -VAE) – showed different divergence choices emphasize different aspects of data (mode-seeking vs coverage). *Amari’s α -connections* – all α -divergences yield the same Fisher metric (second-order), hence “ α -neutral” in that sense ²⁶, but first-order effects differ.

We add a *new twist*: using α specifically to probe model completeness. We propose that varying α changes the “stance” on what counts as a model violation – e.g. $\alpha=1$ (KL) heavily penalizes small probability mismatches, while $\alpha=0$ (Hellinger) is more forgiving of tail differences ^{51 52}. By treating α as a dial, we can observe how the *feasibility of finding a Markov completion* changes. For instance, maybe under a relaxed divergence (say α slightly less than 1), an approximate semigroup exists (the data is “almost” Markov), whereas under strict KL it doesn’t – this would suggest the system is nearly Markov except for outliers. We also

We assume that the divergence measure can be *chosen by the modeler* and is not sacrosanct – that’s a departure from classical statistical inference which usually defaults to KL ($\alpha=1$) or another fixed criterion. In doing so, we implicitly assume the availability of enough data to compute different divergence measures reliably (since some, like α far from 0 or 1, might emphasize tails that need lots of data). We also assume that the choice of α does not change the underlying “truth” but changes our perspective on it – an epistemic rather than ontic view. Prior art in robust stats assumes one true distribution and

To validate this, we can conduct experiments where we know a system’s degree of Markovianity and see if different α values detect it. For example, generate data from a process that has slight deviations from Markov property (e.g. a mixture of Markov chains or a Markov chain with occasional outlier transitions). Then attempt Markov completion under various α : perhaps KL ($\alpha=1$) fails the compatibility test (data looks inconsistent), but $\alpha=0.5$ succeeds (ignores the outliers). We’d observe how the “gap” in divergence behaves – if D_{α} is minimal at some α , that might indicate the appropriate stance to treat the system as Markov “for practical purposes.” Another approach: use synthetic noise added to transition probabilities and show that as α varies, the neutral operator’s

Low. While robust inference literature covers α in estimation, no one has explicitly tied it to *whether a dynamical model exists*. The risk is that some might consider this just an application of robust estimation (“you’re just fitting with different divergence measures”). We differentiate by emphasizing that our *goal* is different: not fitting per se, but testing model viability and stability. Since we are introducing new terminology (like “stance”) and new uses (diagnostics for α), it’s unlikely to clash with prior claims – it’s more likely to be seen as an interesting extension of robust methods into a new domain.

suggest practical
**diagnostics for
choosing α :** e.g.
trying different α
and seeing
which yields
more stable or
simpler
completion
might inform an
analyst whether
to prioritize
fidelity or
robustness ⁵¹

⁶³ . This is novel
because typically
 α is fixed; here it
becomes an
investigative
tool. Moreover,
we link α to
**completion
stability:** the
idea that some
systems might
only admit a
completion in
the limit $\alpha \rightarrow 0$
(very forgiving,
essentially
focusing on
support overlap),
indicating a
highly unstable
completion
where small
changes in
distribution
matter.
Conversely, if a
completion
exists even for
 $\alpha=1$, it's very
stable/robust.
This gradation of
completion

picks α to
estimate it; we
assume an
underlying
process may or
may not exist
and pick α to
test hypotheses
about its
existence. This
is a subtle but
important shift
in assumption:
divergence is a
tool for *model
existence testing*,
not just
parameter
estimation.

estimated
generator
converges or
diverges. If our
theory is right,
there should be an
 α that yields a
stable semigroup
approximation for
a mildly non-
Markov system.
These experiments
would demonstrate
that α is not
just a theoretical
knob but has
tangible effects on
model inference.

Novelty Claim	Closest Prior Art	What's New Here	Different Assumptions	How to Test / Validate	Risk of Overlap
		quality via α is a new concept.			

(v) IG-leaf invariants (comparability across factorizations)

– Defining properties (especially spectral features) that remain the same for all interior completions producing the same boundary data, enabling model-agnostic comparisons.

Markov chain lumpability and spectral theory (Kemeny & Snell 1976) – some eigenvalues are invariant under state aggregation. *Linear system isomorphism* – minimal realizations share transfer function (hence poles/zeros invariants). *Computational mechanics* (Shalizi, Crutchfield) – the entropy rate and statistical complexity are invariants of the process, independent of representation. Recent *Koopman mode analyses* – delay embedding can retrieve true eigenvalues of the system without explicit model ¹⁴.

Our framework posits that if two models lie in the **same “IG leaf”** (meaning they give the same one-step and perhaps multi-step observable distributions), then certain quantities – notably the spectrum of the associated semigroup (eigenvalues of the neutral operator, continuous spectrum shape, etc.) – are **invariant** ⁴⁷. In other words, observational equivalence implies spectral equivalence (at least up to some notion of conjugacy or common invariant subspaces). This is a bold hypothesis that generalizes known special cases: for example, any two hidden Markov models generating the same output process will have the same *output*

We assume that the observational data is rich enough to estimate these invariants (which often means assuming stationarity or ergodicity so that spectral measures make sense). We also assume that models in the same leaf are *spectrally similar in the appropriate representation* (this is part conjecture, part built on heuristic evidence). Prior work would often assume a specific model to compute spectral properties; we assume we can bypass that. In doing so, we also assume an equivalence in the sense of IG: i.e. that two models can be continuously transformed or related by a series of morphisms while staying in

Testing this claim is challenging but feasible in parts. One approach: take two different realizations known to be observationally equivalent (for example, the 2-state vs 3-state HMM that produce the same output statistics as mentioned before). Compute their spectra (e.g. the eigenvalues of the estimated transition matrices or Koopman operators) and show they match. This would empirically support spectral invariance. Another test: for a given complex dataset, fit two very different models (say, one via a delay embedding and one via a hidden-state HMM learned by EM). If our hypothesis is right, certain properties (like the leading eigenvalues of the empirical transition operator, or the slow decay rates in autocorrelation) will agree between the two models, even if their

Moderate-High (conceptually unique). The idea of focusing on invariants is not new (it’s common in physics and in minimal realization theory), but applying it so generally to dynamical systems with partial information is quite new. We have to be careful to note which parts are conjectural. There is little direct prior art claiming “all models in an equivalence class share the same spectrum” – so we’re not overlapping so much as *proposing* it. The risk is someone might point out a counterexample (two different processes with same one-step distribution but different higher-order correlations might have different spectra). If such cases exist, our claim would need refinement (perhaps requiring multi-lag data, not just one-step). We mitigate overlap

autocorrelation structure, thus the same dominant eigenvalues for the output transition operator. We articulate this idea and even outline a result that “within each IG-equivalence class, Koopman generators are resolvent-conjugate,” meaning their spectra coincide ⁵⁸. The novelty is making this precise and leveraging it: we aim to **compute spectral invariants directly from data** (like via Hankel-DMD or other operator approximations) and use them as features of the underlying process that do not depend on which model one chooses. Additionally, we identify other possible invariants on IG leaves, such as information-theoretic quantities (entropy rate,

the same leaf – this ties to the groupoid assumption. Classical approaches either assume a concrete aggregation (lumping) or exact conjugacy. We assume a broader equivalence (IG-equivalence which could include nonlinear transformations preserving information) and still claim invariants exist, which is a stronger assumption but one we argue for in the framework. Some of these invariants (like “IG leaf Fisher curvature”) might be entirely new constructs with no prior definition; we assume they’re well-defined.

internal structures differ. We could also simulate a process and then alter its state representation (e.g. relabel states, or add redundant split states) – these operations shouldn’t change the IG-invariants by design. We then verify our framework’s computed invariants (like an estimated generator’s eigen-spectrum, or an information curvature diagnostic) are indeed unchanged. This would validate that we’re capturing something intrinsic. To test the *conjectural* part (resolvent conjugacy for nonlinear systems), one might need to prove it mathematically for certain classes or use numerics to see that all models fitting the data yield the same spectral lines.

risk by clearly delineating proven invariants (e.g. entropy rate, maybe certain eigenvalues in lumpable cases) versus conjectured ones (full spectral conjugacy). In essence, we’re breaking new ground here, so the bigger risk is unproven correctness rather than duplicating existing work.

multi-
information at
lag k) or
geometric
curvatures in the
information
manifold, as
candidate
invariants. By
mapping key
terms (like
“generator limit,”
“edge Fisher
curvature”) to
known invariants
in other fields,
we give a
dictionary so
researchers can
see that, for
instance, our
“neutral spectral
gap” is
analogous to the
mixing time of
any Markov
realization. The
**comparative
novelty** is that
instead of saying
“if your model is
wrong, you get
wrong
eigenvalues,” we
say “don’t worry
about the model
at all – just
compute the
eigenvalues from
the data; those
are intrinsic to
the process.”
This turns model
non-
identifiability
from a problem
into an

Novelty Claim	Closest Prior Art	What's New Here	Different Assumptions	How to Test / Validate	Risk of Overlap
		opportunity: you focus on what <i>can</i> be identified (the invariants).			

Notes: In the above, each novelty is contrasted with prior work, but many of these contributions are **interdependent** (e.g. the groupoid view (iii) enables defining leaf invariants (v), and the α -divergence stance (iv) ties into the boundary-first ethos (i)). We have highlighted assumptions and validation approaches to clarify which aspects are solid vs. conjectural (e.g. the spectral invariance is presented as a conjecture to be proven or tested, whereas the existence of multiple equivalent models is already known and accepted) ⁵⁸

⁵⁰ .

4. Terminology Alignment Guide

The boundary-first completion framework introduces new terminology or uses terms in specific ways. The following guide maps these key terms to analogous or familiar concepts in existing literature:

- **Boundary data / Edge marginal:** In our framework, an “edge marginal” refers to the empirical **one-step joint distribution** between consecutive observations (and by extension, finite-*lag* joint distributions) ⁶⁴ . This is analogous to an **empirical transition matrix** or **one-step observable distribution** in a Markov chain. In the literature, one might simply call this $P(Y_t, Y_{t+1})$ or the *lag-1 joint*. It’s what HMM theory would derive from observations to estimate transition probabilities, and what time-series analysts compute as the one-lag covariance or frequency matrix. We elevate it to a first-class object. Essentially, *edge marginals* = *observed transition statistics*.
- **Interior / Completion:** The “interior” is the hypothesized dynamical model (Markov chain or semigroup) that *completes* the boundary data. What we call a **completion** is the process of finding a **stochastic process or semigroup** consistent with the observed marginals ¹⁶ . In classical terms, this is akin to the **Kolmogorov extension** or **Markov extension** problem – given finite-dimensional distributions, find an infinite-dimensional one (here with the Markov property). It also relates to **system identification**: inferring a state-space model from data. Our use of “completion” emphasizes it as a closure operation: much like completing a sigma-algebra or closing an equation set. In other words, *Markov completion* = *finding a full Markov process consistent with given partial observations (marginals)*.
- **Neutral operator / Neutral geometry:** The **neutral operator** $N(\Gamma_Y, D_f)$ is a constructed linear operator (acting on functions of Y) that we derive *directly from the one-step empirical distribution and a chosen divergence* ⁶⁵ . It is “neutral” in that it does not assume any particular hidden state structure – it’s defined on the observable space using an f -divergence to weight transitions. In traditional terms, this corresponds to an **observable transition operator** or **empirical Perron-Frobenius/Koopman operator** estimated from data ⁶⁶ ²³ . However, unlike a standard empirical transition matrix (which assumes Y itself follows a Markov chain), the neutral operator can be seen as a *representation-agnostic approximation* to a true Markov operator, if one exists. The **neutral geometry** refers to the information-geometric structure (Fisher metric and dual

connections) induced by treating the space $\mathcal{P}(Y \times Y)$ of edge distributions with a Csiszár–Amari divergence ⁶⁷ ²⁶. In simpler terms, this is just the classical **statistical manifold geometry** on the space of transition probabilities – often one uses the Fisher information metric on transition matrices (as in Markov chain information geometry) ⁶⁸. We call it “neutral” to stress that it doesn’t depend on choosing a particular model parameterization; it’s the natural geometry *any* model must respect on the observational distribution. So, *neutral operator* ~ *empirical transition operator (model-free)*; *neutral geometry* ~ *canonical IG geometry on probability distributions (especially transition laws)*.

- **Completion groupoid:** This refers to the collection of all possible “completions” (Markov models) for a given set of boundary data, equipped with transformations between them (like state renaming or refinement) that form a groupoid structure. In plainer language, this is essentially the **equivalence class of all Hidden Markov Models that give the same observable process**, along with the mappings that convert one model to another without changing the observables. In HMM literature, one talks about different *realizations* being **observationally equivalent** ²¹. Those realizations can often be transformed via **state splitting/merging or relabeling** operations. If an operation is invertible (bijective relabeling), that’s like a groupoid isomorphism (actually a group action on the model set). The term “groupoid” is borrowed from category theory: here each model is an object, and a morphism exists if one model can simulate another’s outputs. Essentially, our completion groupoid is analogous to **Willems’ behavioral equivalence class** in system theory or the set of all **congruent realizations** in control (where different state-space representations have the same input-output behavior). We make it explicit that we consider all these models on equal footing. *In sum: completion groupoid = set of all observationally equivalent models (HMMs) under state transformations*. This concept ties into **lumpability** and **conjugacy** in Markov chains, where one can map one chain to another by aggregating states without changing the output process.
- **IG-equivalence / IG leaves:** Two realizations are **IG-equivalent** if they cannot be distinguished by the information-geometric invariants at the “boundary” level – effectively they lie on the same **IG leaf** ⁴⁶. An **IG leaf** is the set of all models (with possibly different internal states) that share the same observed one-step distribution *and* induce the same Fisher information structure on that distribution. In other words, if you consider the manifold of all possible one-step distributions (and perhaps higher-step if needed), an IG leaf might fix those distributions to specific values. This is analogous to a **fiber** in a fiber bundle: the base space could be the space of observable distributions, and each fiber (leaf) consists of all parameter sets of different models that project to that same distribution ⁴⁸. In literature, one could compare this to **sufficient statistic fibers**: e.g. different parameter values that yield the same likelihood function on observed data form a manifold (the concept of **phenomenological equivalence**). Specifically, for Markov chains, Amari (2001) describes fixing lower-order marginals defines a submanifold ⁴⁵ – that’s essentially the IG leaf we mean. So, *IG leaf = all models with the same “footprint” in observable distribution space*. We use it to analyze invariants: anything true for one model in the leaf is true for all (by definition, at least for the boundary statistics; we conjecture spectral properties too).
- **Edge Fisher metric / Fisher curvature:** Inside the neutral geometry, we have objects like the Fisher information matrix associated with varying the edge distribution. The **edge Fisher metric** is just the **Fisher information metric** on the space of one-step distributions $\mathcal{P}(Y \times Y)$ ³¹. If we consider a parametric family of transition probabilities Γ_θ , the Fisher metric at θ tells us how sensitively Γ changes with θ . In literature, this is exactly what you’d compute for a statistical model of transition matrices (e.g. see **information geometry of Markov chains**,

where one might consider an exponential family of transition matrices and compute its Fisher information ⁶⁸). “Fisher curvature” could refer to certain curvature scalars derived from that metric, indicating model complexity or incompatibility (e.g. non-zero curvature means the family of distributions is not an exponential family or has interactions). In short, *edge Fisher metric = standard Fisher information on transition probabilities; edge curvature = any IG curvature (sectional, Ricci, etc.) evaluated for the manifold of edge distributions*. We mention these because checking curvature can be a way to see if a model lies in a “flat” submanifold (which might correspond to a simpler representation) ¹⁸.

- **Generator limit:** This refers to the **infinitesimal generator** of a continuous-time Markov semigroup, obtained as the limit of the discrete-time transition operator as the time step $\Delta t \rightarrow 0$ ⁶⁹ ¹⁶. In classical terms, if $P(\Delta t)$ is the transition matrix over interval Δt , the generator is $L = \lim_{\Delta t \rightarrow 0} \frac{P(\Delta t) - I}{\Delta t}$ (when it exists), and $e^{L \Delta t} = P(\Delta t)$. In our framework, we talk about a **“generator emerges from limits”** when the empirical family $\{\Gamma_Y(\Delta t)\}_{\Delta t}$ converges to a semigroup ⁷⁰ ⁶¹. This aligns directly with standard **Markov process theory** (the link between discrete and continuous time via Hille–Yosida theorem). So generator limit is just the **Markov process generator** (think of the Q-matrix in a continuous-time chain, or the infinitesimal generator of a C_0 -semigroup in functional analysis). We emphasize it as a *limit of empirical operators* to stress that continuous dynamics are *not assumed a priori* but *obtained by letting Δt shrink* ⁷⁰ ⁶¹. In summary, *generator limit = the infinitesimal generator (e.g. transition rate matrix) you get by taking finer and finer steps if possible*. If the data comes at multiple time scales, this is how we’d detect a true continuous-time dynamic.
- **“Rose” condition / Measure fidelity:** The **Rose condition** is a term we use for the requirement that the divergence between the empirical one-step distribution and that induced by a candidate model is zero ³³. In simpler terms, it means *the candidate model exactly matches the observed one-step marginal*. This is nothing but **consistency at lag-1** or equality of transition probabilities. It’s named in our docs (perhaps as an acronym or homage) and corresponds to what one might call *goodness-of-fit at the one-step level*. In literature, this is analogous to saying the model has the correct sufficient statistic (the correct first-order Markov structure). It’s the basic check an HMM must pass: its implied $P(Y_t, Y_{t+1})$ must equal the empirical Γ_Y . We break fidelity into topological (qualitative), measure (this Rose condition), and geometric (finer Fisher alignment) levels ¹⁷ ¹⁸. So Rose condition = *exact match of one-step distribution (no divergence)* – a prerequisite for any completion. It’s essentially the condition of **“edge equality”**: $D_f(\Gamma_Y \parallel \Gamma_Y^{\text{model}}) = 0$ means $\Gamma_Y^{\text{model}} = \Gamma_Y$ ³³.
- **Neutral (α) stance:** When we say an α -neutral empirical stance, we refer to the idea of choosing a specific α -divergence for the neutral operator. An **α -stance** means we treat discrepancies in a certain way: e.g. $\alpha=1$ stance = I will be sensitive to any likelihood difference (KL), $\alpha=0$ stance = I weigh differences in a more balanced way (Hellinger). This maps to the concept of **power divergence** or **β -divergence** in robust statistics, where different exponents reflect different robustness. It’s analogous to the **“belief” or prior one has about noise**: for example, using an L^2 norm vs L^1 norm in fitting. Here, α is like choosing the norm in probability space. Literature analogs include **M-density approach** in robust stats and **α -VAE** in machine learning. So, neutral α stance = *choice of divergence measure to judge model fit; analogous to choosing different loss functions in estimation (like MLE vs least squares), which reflect one’s stance on outliers vs typical points*.

By aligning these terms, we see that much of our terminology is a re-framing of classical concepts: “edges” = *observed transitions*; “completion” = *model identification*; “neutral geometry” = *Fisher info geometry on observation space*; “groupoid of completions” = *equivalence class of models under output equivalence*; “IG leaves” = *manifold of distributions with fixed lower-order marginals (as in hierarchical modeling by Amari)* ⁴⁵; “generator limit” = *infinitesimal generator of Markov process*. The novelty lies in how we combine and utilize these, rather than in the basic concepts themselves.

5. Priority Reading List (20–30 Key References)

To contextualize and deepen understanding of the boundary-first completion framework, we recommend the following references. They are ordered roughly by relevance, with brief justifications:

1. **Shun-ichi Amari (2001)** – “*Information geometry on hierarchy of probability distributions of Markov chains.*” **Justification:** Directly relevant as it studies manifolds of joint distributions with given lower-order marginals ³⁵. Introduces orthogonal decomposition of dependencies, laying groundwork for IG analysis of dynamical chains. Helps connect our IG-leaf concept to known results by Amari on Markov chains.
2. **D. Blackwell & L. Koopmans (1957)** – “*On the identifiability problem for functions of finite Markov chains.*” **Justification:** A classical result showing non-uniqueness of HMM representations ²¹. Demonstrates that different hidden Markov models can produce the same observable process, motivating our completion groupoid and invariants approach. A foundational read to understand the long-recognized identifiability issues our framework addresses.
3. **Littman, Sutton, and Singh (2002)** – “*Predictive representations of state.*” **Justification:** Introduces PSRs, which inspire our boundary-first stance by constructing state from observable quantities ²⁴. Reading this provides insight into prior data-driven state construction methods, highlighting how our approach extends them (by adding IG structure and explicit closure tests).
4. **B. O. Koopman (1931)** – “*Hamiltonian systems and transformation in Hilbert space.*” **Justification:** The seminal paper introducing the Koopman operator approach ¹. While dated, it’s the foundation of operator-theoretic dynamics which we heavily leverage. Understanding Koopman’s perspective on evolving observables is crucial for our operator-neutral viewpoint.
5. **F. Takens (1981)** – “*Detecting strange attractors in turbulence.*” **Justification:** The original statement of the Takens embedding theorem ¹¹. This is key background for how one can reconstruct dynamics from observations. It contrasts with our probabilistic completion: Takens provides conditions for deterministic reconstruction (topological fidelity), which our framework seeks to augment with measure-theoretic (stochastic) fidelity.
6. **T. Sauer, J. Yorke, and M. Casdagli (1991)** – “*Embedology.*” **Justification:** Extends Takens to more general and fractal attractors. Important to see the robustness and limitations of delay embedding. Our work diverges by focusing on stochastic processes and partial observations, but knowledge of embedology helps frame the differences (e.g. we need additional conditions beyond those in embedology to ensure a Markov process exists).

7. **I. Csiszár (1967)** – *“Information-type measures of difference of probability distributions.”* **Justification:** Introduces f -divergences, the mathematical basis of our α -divergence family. A primary source for understanding the measures (like KL, Hellinger) we use in the neutral operator ²⁶. It provides context on why these measures have certain properties (convexity, etc.) that we exploit in our consistency conditions.

8. **S. Amari & H. Nagaoka (2000)** – *“Methods of Information Geometry.”* **Justification:** An authoritative textbook on information geometry. It covers the Fisher metric, α -connections, and a variety of examples. Particularly relevant for understanding the canonical IG structures we call “neutral geometry,” and the concept of α -neutrality ²⁶. It’s a comprehensive reference for any IG aspects of our framework.

9. **Y. Ohara, T. Tanaka, S. Fujimoto (1995)** – *“Differential geometry of statespace and its application to system and control theory.”* **Justification:** Representative of the 1990s work applying IG to control systems ³⁶. It shows how Fisher information can be defined for state-space models and introduces natural gradients for adaptation. This connects to our use of IG on dynamical models and helps a reader see how prior work handled parametric system families (whereas we do nonparametric).

10. **B. Vanluyten, B. De Moor, J. Willems (2007)** – *“Equivalence of state representations for hidden Markov models.”* **Justification:** Directly addresses when two HMMs are equivalent and how to reduce them ²¹. Reading this gives insight into known algorithms for HMM minimization and equivalence classes, essentially the practical side of our completion groupoid idea. It’s a modern take in line with Willems’ behavioral philosophy, relevant for benchmarking our approach.

11. **Jan C. Willems (1989)** – *“From time series to linear system – Part I. Finite-dimensional linear time-invariant systems.”* **Justification:** Although focused on linear systems, this work lays out the behavioral approach (no a priori state, model = set of trajectories). It’s fundamental to understand the philosophy of model as a *closure* of data patterns, which we adopt and generalize. It also provides an algorithmic angle (via Hankel matrices) that parallels how we might approach building a neutral operator from data.

12. **Herbert Jaeger (2000)** – *“Observable operator models for discrete stochastic time series.”* **Justification:** Jaeger’s work on OOMs is a precursor to PSRs and is very much in spirit with our framework (models determined by observable probabilities). It’s helpful to see the mathematical formulation he uses and the limitations he notes (e.g. negativity issues in OOMs). Understanding OOMs will highlight what improvements our framework seeks (like ensuring a valid probabilistic operator and adding IG constraints).

13. **Byron Boots, Amit Singh, Geoff Gordon (2011)** – *“A spectral learning approach to infinite-horizon prediction.”* **Justification:** This paper presents a spectral method for learning PSRs (using singular value decompositions of Hankel matrices). It’s relevant as it shows how one can learn an implicit model without assuming states, somewhat analogously to our neutral operator approach. It also connects to how spectral invariants (Hankel singular values) relate to system properties. This is state-of-the-art ML approach to sequence modeling which our framework complements with a theoretical correctness angle.

14. **A. Basu, I.R. Harris, N. Hjort, M. Jones (1998)** – “*Robust and efficient estimation by minimizing a density power divergence.*” **Justification:** Introduces the density power divergence (essentially an α -divergence) and discusses the balance of robustness vs efficiency. It is directly relevant to our use of α as a tuning parameter ⁵¹. By reading this, one gains intuition on how changing α down-weights outliers, which parallels how we interpret α -stance in model completion (down-weighting unlikely transitions etc.).

15. **A. Cichocki & S. Amari (2010)** – “*Families of alpha-beta divergences.*” **Justification:** A survey that generalizes and discusses various divergence families. This is useful to understand the landscape of possible divergence choices beyond just KL and Hellinger. It reinforces the idea that a continuum of divergences exists and they have different properties – the underpinning of our stance parameter. It’s more of a reference to see the formulas and known invariances (like joint convexity, etc.) that justify why all α in $[-1,1]$ are “neutral” in inducing the same Fisher metric ³¹.

16. **Igor Mezić (2013)** – “*Analysis of fluid flows via spectral properties of the Koopman operator.*” (*Annual Review of Fluid Mechanics*). **Justification:** A modern survey of Koopman operator theory and its applications. This gives a broad perspective on why spectral analysis of the Koopman operator is powerful ⁷ ⁷¹. It will help readers appreciate what our framework gains by achieving a Koopman representation *if* a completion exists (and what is lost if it doesn’t). The paper also touches on the need for coherent structures which ties into eigenfunctions – an underpinning for our spectral invariants idea.

17. **M. Budisić, R. Mohr, I. Mezić (2012)** – “*Applied Koopmanism.*” (*Chaos journal*) **Justification:** An accessible tutorial on applying Koopman operator analysis to data. It includes discussions of practical computation (DMD) and examples. Useful for understanding how one goes from data to an approximate operator, analogous to how our neutral operator is computed. It also emphasizes linear representations of nonlinear dynamics, reinforcing why having a Markov completion (thus a linear operator on observables) is beneficial.

18. **Arbabi & Mezić (2017)** – “*Ergodic theory, Dynamic Mode Decomposition and computation of spectral properties of the Koopman operator.*” **Justification:** This work proved that by stacking delay coordinates (Hankel matrices) and doing DMD, one can recover the true Koopman spectrum for ergodic systems ¹⁴. It’s a key result linking the delay embedding world and operator spectra. It provides theoretical support for our usage of spectral invariants from data – essentially showing that under certain conditions, data-driven spectral analysis is valid. This helps justify our IG-leaf spectral conjecture, at least in the deterministic ergodic case, by analogy.

19. **J.G. Kemeny & J.L. Snell (1976)** – “*Finite Markov Chains.*” **Justification:** This classic text (or their earlier papers) covers Markov chain theory including *lumpability*, *stationary distributions*, and *eigenvalues*. Notably, it discusses when you can aggregate states without changing the chain’s behavior. Chapter sections on lumpability or invariant subspaces align with our notion of different models having shared spectra. It’s good foundational reading to recall known invariants in Markov chains (like Kemeny’s constant, or that the subdominant eigenvalue is the same in a lumped chain if lumping is proper).

20. **J.P. Crutchfield & K. Young (1989)** – “*Inferring statistical complexity.*” **Justification:** A landmark paper in computational mechanics, introducing the concept of statistical complexity and ϵ -

machines. It formalizes how to construct a minimal predictive model from data and defines invariants like entropy rate and statistical complexity that are properties of the process itself. This is closely related to our goals: *statistical complexity* is an invariant (the size of the minimal model) and *entropy rate* is an invariant (the process's unpredictability). Reading this provides a deeper theoretical background for data-driven model inference and underscores the idea of focusing on invariants of the process (not of any specific model's states). It will resonate with our IG-leaf invariant approach, though from a different angle (algorithmic complexity vs. geometric invariants).

21. **C.R. Shalizi & J.P. Crutchfield (2001)** – “*Computational Mechanics: Pattern and Prediction, Structure and Simplicity.*” **Justification:** Follow-up work that elaborates the computational mechanics framework and discusses causal state equivalence in detail. It’s beneficial for understanding the rigorous way of defining when two histories are equivalent (predictively), which parallels our use of divergence to define equivalences. It also discusses when additional structure in the process (higher-order memory) increases complexity – analogous to when our completion fails (needing higher-order edges). Helps in appreciating the theoretical limits of predictive modeling.

22. **Y. Ephraim & N. Merhav (2002)** – “*Hidden Markov processes.*” (*IEEE Trans. Information Theory*, survey) **Justification:** An extensive survey of HMMs and related inference problems. It covers filtering, estimation, as well as issues like mixing and even mentions identifiability challenges. This is a good all-in-one reference for classical stochastic process inference. It’s relevant as background to see what mainstream approaches do and don’t do – for instance, it doesn’t discuss testing for the existence of an HMM, it assumes one. This contrast will highlight the niche our framework fills. It’s also useful if one wants to see how far one can get with traditional tools (for example, there’s discussion of the EM algorithm’s landscape, which implicitly relates to uniqueness of solutions).

23. **Kolmogorov (1933)** – “*Foundations of the Theory of Probability.*” (especially the Extension Theorem) **Justification:** This is the foundational theorem about extending consistent finite-dimensional distributions to an infinite probability space. While abstract, it provides the measure-theoretic underpinning for any completion problem. Our framework can be seen as a specialized application plus extra constraints (Markov property) of Kolmogorov’s extension. Reading a summary or explanation of this theorem (e.g. in a measure-theoretic probability text) will give rigorous conditions required (consistency, etc.) and assure the reader that what we ask (consistency across lags) is rooted in established theory.

24. **Israel, Rosenthal, & Wei (2001)** – “*Finding Generators for Markov Chains via Empirical Transition Matrices.*” **Justification:** This paper addresses the “embedding problem” for Markov chains: when does a given discrete-time transition matrix have a continuous-time generator? It’s very relevant for the continuous-time aspect of our framework (generator limit). It lists criteria and counterexamples for embeddability. This informs our continuous-time completion conditions ⁷⁰ ⁶¹. By reading it, one understands the non-trivial nature of going from $P(\Delta t)$ to L and that not every matrix is embeddable – reinforcing why our limiting procedure and Hille–Yosida conditions matter.

25. **P. Gaspard & X.-J. Wang (1993)** – “*Noise, chaos, and (ϵ, τ) -entropy per unit time.*” **Justification:** (An example from dynamical systems literature bridging deterministic chaos and stochastic processes.) It discusses entropy rates and complexities for processes with noise and chaos. Included here to give a physical perspective on why one might want a framework that can handle both deterministic-like behavior and stochastic features. It’s tangentially related but helps broaden understanding of the

context where neither purely mechanistic nor purely stochastic approaches suffice, much like the motivations of our framework. It underscores the need for tools that can quantify uncertainty and structure (like our IG invariants) in complex systems.

26. **Nihat Ay, et al. (2017)** – *“Information Theory and Interaction: A Joint Perspective on Complex Systems.”*
Justification: (If available: this book or related papers by Ay.) Ay has worked on information-theoretic approaches to complex systems (e.g. multi-information decomposition in time series). His perspectives connect information geometry, information theory, and complex system structure. We include this to show the conceptual adjacent work – it doesn’t directly solve our problem but provides a way of thinking about complexity, emergence, and the role of information constraints. It resonates with our emphasis that “irreducible pluralism” (multiple models or none) must be acknowledged ⁶. It helps frame the philosophical underpinning of why our stance is needed in complex systems science.

27. **T. Schreiber (2000)** – *“Measuring information transfer.”* **Justification:** Introduces transfer entropy, a model-free statistic to detect dynamical causal influence. While not explicitly used in our framework, it’s an example of an *observable-only* measure of dynamics. It aligns with our boundary-first ethos by extracting insight (directional information flow) without assuming a model. It’s a short, insightful read showing how much can be done with data alone, setting a precedent for our aim to derive invariants and partial structure empirically.

28. **A. Chentsov (Čencov) (1972)** – *“Statistical Decision Rules and Optimal Inference.”* **Justification:** (For completeness in IG theory.) This is the original source (in Russian, later translated) of the theorem characterizing the Fisher metric as unique under Markov maps ²⁸. We list it for those interested in the deep mathematical foundation of information geometry, which justifies why we focus on Fisher as “canonical.” It’s highly theoretical, so it’s lower priority unless one wants to delve into the underpinning of α -neutrality.

29. **E. N. Lorenz (1963)** – *“Deterministic nonperiodic flow.”* **Justification:** (Though not directly cited above, Lorenz is a classic example in dynamical systems.) We include it as a reminder of how deterministic chaos taught us that even with known equations, long-term prediction is limited – motivating why one might choose a boundary-first empirical approach. It’s largely historical context: reading Lorenz’s paper highlights the unpredictability in simple mechanistic models, strengthening the argument for empirically driven frameworks in complex systems.

30. **G. Birkhoff (1931)** – *“Proof of the ergodic theorem.”* **Justification:** Another foundational piece, included for theoretical completeness. Ergodic theory underlies many assumptions in our framework (e.g. that time averages equal ensemble averages, which is why empirical distributions make sense for underlying processes). Birkhoff’s theorem ensures that if a completion (Markov chain) exists and is ergodic, our empirical estimates converge. While not necessary to implement our framework, an understanding of ergodicity (via Birkhoff) is intellectually satisfying and assures that our empirical approach has a basis when the process is well-behaved.

(The above list blends classical and modern, theoretical and practical. The top half-dozen references are the most directly illuminating for this framework’s unique contributions, while the rest provide important background and adjacent perspectives. Together they give a 360° view of the landscape in which the boundary-first completion program is situated.)

6. Potential Reviewers and Relevant Communities

Identifying the right audience is crucial. Below is a list of researchers and communities who would be well-suited to appreciate, review, or provide venues for this work, each with a note on their connection:

- **Shun-ichi Amari (RIKEN Brain Science Institute, emeritus)** – *Information Geometry pioneer*. Amari would appreciate the use of α -divergences, Fisher metric, and the idea of an information-geometric foliation of model space. He has worked on statistical manifolds of time-series and would recognize the novelty in applying IG so thoroughly to dynamics. (As a senior figure, he might not actively review, but his intellectual lineage – e.g. former colleagues/students – certainly fits.)
- **Nihat Ay (Leipzig & Santa Fe Institute)** – *Complex systems & information theory*. Ay has bridged IG, information theory, and dynamics (e.g. studying graphical models, complexity measures in time-series). He would be an ideal reviewer who understands both the theoretical IG aspects and the practical implications for complex systems. His community (e.g. the Entropy and Information Theory in Dynamical Systems group) would find the “boundary-first” stance intriguing as it aligns with objective measures of complexity.
- **Frank Nielsen (École Polytechnique & Sony CSL)** – *Information geometry and divergence expert*. Nielsen has published extensively on different divergences and their properties. He would be interested in our novel use of the α -divergence family as a tuning parameter. As someone active in the Geometric Science of Information (GSI) community, he could evaluate both the IG rigor and the empirical implications.
- **Igor Mezić (UCSB)** – *Leader in Koopman operator theory*. Mezić’s work on spectral methods for dynamics is directly relevant. He would be interested in the conditions for existence of a semigroup from data and the idea of multiple systems sharing a Koopman spectrum. As a reviewer, he could assess our operator-theoretic claims (like the resolvent conjugacy conjecture) and would appreciate the unifying of embedding and spectral viewpoints. His group or the SIAM Dynamical Systems community he engages with would be a prime audience for the spectral parts of our work.
- **Steven L. Brunton (University of Washington)** – *Data-driven dynamics and control*. Brunton works on DMD, Koopman, and machine learning for dynamical systems. He has an interest in bridging physical modeling with data science. He would likely appreciate the framework’s attempt to systematically identify when a dynamic model is reliable. As an advocate of combining delay embeddings with modern Koopman analysis, he’d align with our approach of looking at multi-scale data and spectral diagnostics ⁹ ¹⁴ . He could also advise on how to communicate this to the data-driven modeling community (e.g. NeurIPS or SciML audiences).
- **James P. Crutchfield (UC Davis)** – *Computational mechanics and complex systems*. Crutchfield has long championed building models (ϵ -machines) from data and focusing on intrinsic computation. He would find the boundary-first philosophy congenial, as it resonates with not assuming more than necessary. As a reviewer, he’d scrutinize our claims about invariants and minimality, given his expertise in causal states. The nonlinear dynamics and chaos community (like the Chaos journal circle or Santa Fe Institute network) to which he belongs would be very interested in the irreducible underdetermination aspect and the hierarchy of model fidelities ¹⁸ .

- **Cosma Rohilla Shalizi (Carnegie Mellon University)** – *Statistical learning and causal architecture*. Shalizi (a student of Crutchfield) extended computational mechanics and worked on causal state splitting/reduction algorithms. He would be a good fit to review the statistical rigor of our approach to discovering models from data. He'd particularly appreciate the admission of non-uniqueness and emphasis on what can be learned if a process isn't nicely Markov (since he studied processes with causal representations of varying complexity). His perspective straddles statistics and physics, much like our framework.
- **Satinder Singh (University of Michigan)** – *Reinforcement learning and PSRs*. Singh co-introduced PSRs ²⁴. He would be intrigued by our extension of the PSR idea with an IG and divergence twist. As someone in machine learning, he could assess how our method might be implemented or tested on real sequence data, and he'd value the shift of treating model existence as conditional. The theoretical ML sequence-modeling community (including Singh, Michael Littman, and others in predictive modeling) is a natural audience for the "empirical stance" and groupoid concept, since they deal with representation learning for partially observable processes.
- **Herbert Jaeger (Jacobs University)** – *Observable operator models and reservoir computing*. Jaeger's OOMs are conceptually very close to our neutral operator idea. He also later worked on reservoir computing, which is another way to encode dynamics without explicit state models. Jaeger would be a great reviewer for the linear algebra consistency and could provide insight into how our divergence criteria relate to OOM conditions (like positivity of learned operators). His community in time-series analysis and unconventional computing would find the idea of a "neutral" stance appealing – it's philosophically aligned with not committing to a particular state interpretation.
- **Lennart Ljung (Linköping University)** – *System identification (classical)*. Ljung is an authority on identification theory for linear and nonlinear systems. While our framework goes beyond traditional system ID, a person like Ljung (or his colleagues in the IEEE Control Systems/Identification community) would be valuable to ensure we properly situate our work relative to known identification algorithms. They would also be potential reviewers or audience especially for the continuous-time aspects and the emphasis on conditions for model existence (since control theory cares about realizability and minimality too). A community like the IFAC System Identification symposium or IEEE Transactions on Automatic Control readership could be interested if we frame parts of this as "when is a Markov model identifiable from data?"
- **Tryphon Georgiou (UC Irvine)** – *Control theory and optimal transport in dynamics*. Georgiou has worked on topics like connecting control, probability, and geometry (e.g. covariance completion problems, Wasserstein in control). He might appreciate the formalization of completion as a type of constrained realization problem. Also, his recent work on using optimal transport metrics for comparing power spectral densities resonates with our use of divergences for comparing processes. As a reviewer, he could comment on our continuous-time limit theorems and suggest connections to convex optimization approaches in system realization.
- **Peter Schuster or Jürgen Jost** (representing theoretical biology/physics community) – *Complex systems and theoretical biology*. Researchers like Schuster (known for sequence to dynamics models in biology) or Jost (who works on networks and information geometry) could provide perspective from complex systems biology or network dynamics. They would be interested in how the framework handles underdetermination, as these fields often deal with many possible models fitting limited

data. They might serve better as audience members (e.g. at a conference like “**Geometric Science of Information (GSI)**” or “**Information Theory and Life Sciences**” workshop), where this interdisciplinary approach would spark discussion.

- **Geometric Science of Information (GSI) community** – *Conference and journal audience*. This community, including people like Frank Nielsen (mentioned), Frédéric Barbaresco, etc., is explicitly focused on information geometry and its applications. They would be appropriate reviewers/venues because our work “squarely lies in that scope, applying IG to a novel domain” ⁷². They would appreciate the new use of α -divergences and the IG foliation idea. A reviewer from this community would ensure our IG terminology and claims are sound.
- **SIAM Dynamical Systems & Ergodic Theory researchers** – e.g. **Brian Hunt, Erik Bollt, or Konstantin Mischaikow**. These are people familiar with both the rigors of dynamical systems and data-driven methods. They could evaluate the mathematical aspects of our semigroup existence criteria and spectral conjectures. The **SIAM Journal on Applied Dynamical Systems (SIADS)** or **Chaos** would likely draw on such reviewers, who value that we connect to Koopman spectra and embedding theorems.
- **Machine Learning (sequence modeling) community** – e.g. **Neil Lawrence, Zoubin Ghahramani (probabilistic modeling), or Le Song (kernel methods for sequences)**. Such researchers understand probabilistic sequence models and modern approaches like RNNs, but also appreciate principled frameworks. They might review this from the perspective of “is this useful for machine learning representation learning?” They would relate to the PSR aspect and the divergence-as-loss-function aspect. A venue like NeurIPS or ICML (for the more theoretical paper) might have reviewers of this type, who can help gauge if our framework can inspire new algorithms or at least provides theoretical limits for ML models.
- **Control and Signal Processing community** – e.g. **Hidenori Kimura or Prasad Dhondt (if active)**, or younger researchers like **Hashimoto, Hayashi** in control IG. They’ve extended information geometry to control and filtering ³⁶ ⁴². They would be interested in how our approach could apply to filtering (treating filtering as an IG projection ⁴²) or model reduction. They might review for a journal like IEEE Transactions on Automatic Control, checking that our continuous-time results align with known Markov process theory and that our approach to non-uniqueness is practically meaningful.
- **Santa Fe Institute and Complex Systems community** – This is more of a broad community (including many mentioned: Crutchfield, Ay, etc.). If targeting a venue like *Complexity* or *Entropy*, these would be the readers. They care about emergent behavior, and terms like “irreducible pluralism when completion fails” ⁶ speak their language. A reviewer here (like Jessica Flack or David Wolpert, who work on the limits of inference in complex systems) would assess our philosophical claims and ensure we’re consistent with complexity science insights.

Each of the above individuals or groups connects with a facet of our work: information geometry, operator theory, machine learning, control, or complexity science. In terms of **venues**, based on this list, likely targets include **the Information Geometry journal** (where IG experts would review) ⁷², **SIAM Journal on Applied Dynamical Systems or Chaos** (where dynamical systems experts like Mezić or Crutchfield might review) ⁷³, **IEEE Transactions on Automatic Control** (for the control/identification angle, with reviewers

from that community) ⁷⁴ , or top-tier ML conferences (NeurIPS) for which one would want reviewers like Singh or Brunton who bridge theory and practice in sequence modeling ⁷³ . We believe engaging these diverse but intersecting communities will ensure the work is rigorously vetted from all angles and finds the audience that can build on its contributions.

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Geometry & Dynamical Systems_ Literature and Framework Mapping.pdf

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19 On the Identifiability Problem for Functions of Finite Markov Chains

<https://www.semanticscholar.org/paper/On-the-Identifiability-Problem-for-Functions-of-Blackwell-Koopmans/7681d53f86cbfbd3508cb747af7d2a096b6c60c5>

56 [PS] Equivalence and Reduction of Hidden Markov Models - DSpace@MIT

<https://dspace.mit.edu/bitstream/handle/1721.1/6801/AITR-1370.ps.Z?sequence=1&isAllowed=y>